MIL-C-82632A(OS)
28 November 1983
Superseding
MIL-C-82632(OS)
3 September 1976

MILITARY SPECIFICATION

CONNECTOR KIT, SPLICE, SINGLE WIRE

This specifIcation Is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification establishes the minimum requirements for the procurement of a single wire splice connector, DL 5545977. The connector joins the shipboard guidance wire to the torpedo guidance wire and maintains electrical continuity in seawater to a pressure of 1500 psig. The connector will accept wire sizes from AWG 25 to AWG 23 and is of one type and class.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified (see 6.2), the following specifications and standards of the Issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified In the solicitation, form a part of this specification to the extent specified heretin.

SPECIFICATIONS

FEDERAL

PPP-B-636	Boxes, Shipping, Fiberboard
PPP-T-76	Tape, Preasure-Sensitive, Adhesive Paper (for Carton Sealing)

MILITARY

MIL-P-116	Preservation-Packaging, Methods of
MIL-W-82599	Wire, Insulated, Hard Drawn Alloy of Copper-Cadmium, No. 23 AWG

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer, Naval Ordnance Station, Standardization/Documentation Divison (Code 524), Indian Head, MD 20640 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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STANDARDS

MILITARY

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage

2.1.2 Other Government documents, drawings, and publications. The following other Government documents form a part of this specification to the extent specified herein.

DRAWINGS

NAVAL SEA SYSTEMS COMMAND

DL 5545977	Data List for Wire Splice Klt for Torpedo Mk 48
DWG 697026	Crimp Tool Assembly
DWG 3017915	Wire, Electrical
DWG 5545977	Wire Splice Kit for Torpedo Mk 48
DWG 5545979	Contact Assembly
DWG 5545980	Pin, Contact
DWG 5545983	Housing, Male
DWG 5545984	Housing, Female

(Copies of specifications and standards required by manufacturers in connection with specific. acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.1.3 Order of Precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this speclfic-ation shall take precedence.

3.1 REQUIREMENTS

3.1 Description The single wire splice connector kit (DL 554977) consists of a pin contact plug packaged with a female housing, and a contact socket assembly packaged with a male housing. When assembled, the famale and male housing constitute the connector which provides a pressure-tight electrical connection of two wires.

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3.2 First article. Unless otherwise specified In the contract or purchase order, preproduction models of the connectors shall be manufactured by the contractor using the methods and procedures proposed for the production lot. These nits shall be tested as specified In Section 4 herein and are for the purpose of determining, prior to starting production, that the contractor's production methods are capable of producing connectors that comply with the technical requirements of the contract (see 4.2.1).

3.3 Physical requirements

3.3.1 Materials. Unless otherwise specified in the contract, requisition, or order, all materials and purchased parts entering Into the final assembly of the connector shall conform strictly to the detailed requirements Indicated an the applicable drawings and specifications listed in DL 554977.

3.3.2 Finishes, coatings construction, and dimensions. Finishes, coatings, construction, and dimensions for the connector shall be strictly In accordance with the applicable drawings and specifications listed in DL 554977.

3.3.3 Marking. Marking of assemblies, subassemblies and parts shall be in accordance with the applicable drawings listed in DL 5545977.

3.4 Environmental requirements

3.4.1 Temperature. The connector shall be capable of withstanding temperatures ranging from -65 degrees to +160 degrees Fahrenheit (F) and thereafter shall show no damage and shall meet all the performance requirements and product characteristics of this specification.

3.4.2 Mositure, fungus, and corrosion. The connector shall be rendered mositure, fungus, and corrosion resistant in accordance with the detail requirements specified in the drawings and specifications listed in DL 5545977.

3.4.3 Hydrostatic pressure. The connector shall maintain a watertight seal and shall maintain an insulation resistance between the inner conductor and the seawater of 100 megohms minimum at 500 volts direct current (vdc) while under seawater pressure from zero to 1500 pounds per square inch gage (psig).

3.5 Performance requirements and product characteristics

3.5.1 Connector mating force. The force required to mate the two connector halves shall not exceed 5 pounds.

3.5.2 Connector unmating force. The connector shall withstand a tensile pull equal to 26.4 ±0.2 pounds for #23 AWG or 20.8 ±0.2 pounds for #24 AWG for a period of 30 seconds without interruption of electrical continuity.

3.5.3 Contact voltage drop. Contact voltage drop shall not exceed the following values:

Test current Amps, DC	Maximum Voltage Drop (Millivolts)	
	Initial	After 30-Day Aging
1	750	1000

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3.5.4 Crimp tensile strength. Contacts crimped to wire sizes provided (see 4.1.2) shall withstand a force of 26.4 ±0.2 pounds for #23 AWG or 20.8 ±0.2 pounds for #24 AWG for a period of not less than 30-seconds.

3.6 Workmanship. The connector shall remanufactured in a thoroughly workmanlike manner and shall be free of defects that affect life, serviceability, or appearance. The standards of workmanship exhibited in any approved first article sample, subject to any qualification stated in the Government's notice of approval, shall be determinative of the requirements of the contract relative to workmanship insofar as not specifically covered by applicable specifications.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for Inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the Inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Classification of inspections. Inspection of the connectors shall be classified as follows:

(a) First article inspection (see 4.2.1).

(b) Quality conformance inspection (see 4.2.2).

4.1.2 Special Equipment. The following items are required by this specification and will be furnished by the procuring activity:

(a) One crimp tool per Dwg 697026.

(b) As required size AWG 23 copper-cadmium wire in accordance with MIL-W-82599.

(c) As required, size AWG 24 copper-cadmium wire In accordance with Dwg 3017915.

4.2 Sampling

4.2.1 Preproduction model

4.2.1.1 Supply requirements. Preproduction models of the item components shall be supplied by the contractor In the quantities shown in Table I. The models shall be numbered as Indicated. The connectors shall be manufactured by the same procedures and methods and at the same plant location proposed by the contractor. for the execution of the contract. The model shall be submitted to the procuring activity for purposes of determining compliance with the requirements of the contract or order and the applicable specifications. Further production of the

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item by the contracor prior to the approval of the procuring activity or completion of inspection and tests on the preproduction model shall be at the contractors risk. Accepted preproduction models will become the property of the procuring activity and will be included in the quantity of connectors called for in the contract or order.

TABLE 1. Identification of samples.

Nomenclature	NAVSEA Drawing	Quantity	Sample Identification Number
Housing, Female	5545984	45	1F to 45F
Pin, Contact	5545980	90	1P to 90P
Housing, Male	5545983	45	1M to 45M
Contact Assembly	5545979	90	1S to 90S

4.2.1.2 First article inspection. The preproduction model, selected in accordance with 4.2.1 and Table I, shall be subjected to the impactions and tests of Table II in the sequence listed. When specified in the contract or purchase order, certified Inspection data covering the results of first article inspections and tests shall be forwarded to the procuring activity.

TABLE II. First article test sequence

Paragraph Number	Test	Samples to be Tested
4.3	Examination of Product	All
4.3.1	Dimensional Measurements, Housings	31F - 45F 31M - 45M
4.4.2	Temperature Cycling (28 days)	1 thru 30 (P,F,S, & M)
4.5.1	Connector Mating Force	1 thru 30 (P,F,S, & M)
4.4.3	Hydrostatic Pressure Test	1 thru 30 (Mated)
4.5.2	Connector Unmating Force	1 thru 30 (Mated)
4.5.3	Contact Voltage Drop	31P to 60P 31S to 60S
4.5.4	Crimp Tensile Strength	61P to 90P 61S to 90S

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4.2.2 Quality conformance sample

4.2.2.1 Inspection lot. An Inspection lot for quality conformance inspection shall consist of 30 each of the connecters and parts which are part of a homogeneous production and are submitted to the designated inspection activity. Homogeneous production is difined as those parts (housings) manufactured from a given set of core pins. Should the core pins have to be replaced, this constitutes the end of one homogeneous production and the start of another homogeneous production.

4.2.2.2 Sampling Connector and part sampling shall be in'accordance with MIL-STD-105, Inspection Level II. The acceptable quality level (AQL) for connectors and parts shall be 1.0 for major characteristics and 2.5 for minor characteristics or as otherwise indicated on component drawings. The total housing samples shall consist, as near as possible, of an equal number of housings from each mold cavity.

4.2.2.3 Testing. The quality conformance sample shall be subjected to the tests of Table III in the sequence listed. When specified in the contract or purchase order, certified inspection data covering the results of quality conformance inspection and tests shall be forwarded to the procuring activity.

TABLE III Quality conformance test sequence.

Paragraph Number	Test	Samples to be Tested
4.4.2	Temperature Cycling	1 thru 30 (P,F,S, & M)
4.5.1	Connector Mating Force	1 thru 30 (P,F,S, & M)
4.5.3(b),(d)	Contact Voltage Drop	1 thru 30 (Mated)
4.4.3	Hydrostatic Pressure Test	1 thru 30 (Mated)
4.5.2	Connector Unmating Force	1 thru 30 (Mated)

4.3 Examination of physical attributes. Each connector and part sampled shall be examined to verify that the materials, design, construction, physical dimensions (with the exception of hardness tests and those dimensions on the male and female housings listed an the applicable Governmaut drawings as a "Major Classification of Characteristics") configurations, markings, and workmanship are In accordance with 3.3.1, 3.3.2, 3.3.3, 3.4.2 and 3.6.

4.3.1 Housing sample disposition. First article samples 31 through 45 (F & M), which have successfully passed the inspection procedures of 4.3, shall be subjected to the following test:

4.3.1.1 Dimensional measurements. Those dtiensions listed as a "Major Classification of Characteristics" on the applicable Government drawings shall be examined on the male and female housings to verify that the housings meet the requirements of 3.3.2 (destruction of the housings will be necessary).

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4.4 Environmental tests. Unless otherwise specified herein, all tests shall be performed under the following conditions:

(a) Temperature: 68 to 86 degrees Fahrenheit (F).

(b) Relative humidity: 30 to 80 percent.

(c) Barometic pressure: 24 to 31 inches of mercury.

4.4.1 Crimping. Crimp 15 contact pairs (15 of each pin contact plug and contact. socket assembly) from first article or quality conformance sample (1-30 P&S) to each of the two wire sizes provided (4.1.2). After crimping the contacts to the wires, seat the contacts in their respective housings. Do not mate the connectors at this time. The following procedure shall be used to connect the contacts to the wires:

(a) Clean ends of wires to be crimped with any commercial liquid hand soap.

(b) Place the connector half in the crimp tool, seating the connector half shoulder against the locator In the tool.

(c) Insert the wire end (do not strip wire insulation) into the connector half until it bottoms fully and is visible through the bleeder hole.

(d) Squeeze the crimp tool handles until the ratchet release feature allows to tool to be opened, and remove the crimped connection.

4.4.2 Temperature cycling (28 days) .

(a) The connectors prepared above shall be subjected to 20 temperature- . cycles consisting of: room temperature to -65 degrees F within one hour, maintaining at -65 degrees F for no less than 6 hours; -65 degrees F to +160 degrees F within one hour, maintain at +160 degrees F for no less than 14 hours and +160 degrees to room temperature within 2 hours. The connectors shall be kept at room temperature for at least 2 hours between cycles. This test will cover a span of approximately 28 calendar days and meet the requirements of 3.4.1.

(b) Following the temperature cycling test, the first article sample shall be subjected to the tests of 4.5.1, 4.4.3, and 4.5.2 in that order. The quality conformance sample shall be subjected to the tests of 4.5.1, 4.5.3(b), 4.5.3(d), 4.4.3, and 4.5.2 in that order.

4.4.3 Hydrostatic pressure test. The mated connector assemblies shall be mounted in a hydrostatic pressure tank with both exposed wire ends of each brought out of the tank through suitable sealing fixtures (more than one connector may be tested at one time). The connectors shall then be subjected to a hydrostatic pressure of 1500 ± 50 psig, for a period of 2 hours and shall meet the requirements of 3.4.3 (Figure 1 shows a typical high-pressure test fixture).

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4.5 Performance and product characteristics tests.

4.5.1 Connector mating force. Securely mount one of the connector halves in a movable holding device capable of measuring the required engaging force. The remaining connector half shall be engaged by hand as in service. The force required-to mate the connector halves shall meet the requirements of 3.5.1.

4.5.2 Connector unmating force.

(a) The wire leads of. the mated connector shall be secured to the jaws of a tensile tested.

(b) A force of 26.4 ± 0.2 pounds for #23 AWG or 20.8 ± 0.2 pounds for #24 AWG shall be applied for 30 seconds to the wires at each end of the connector, tending to pull the connector halves apart.

(c) This force shall be applied with 100 milliamps flowing through the test sample and a discontinuity monitor connected to check for any circuit opens which might occur. The connector shall meet the requirements of 3.5.2.

(d) The tensile force shall then be Increased to destruction and the force and type of destruction recorded.

4.5.3 Contact voltage drop.

(a) Crimp 30 contact pairs (each pair consisting of one pin contact plug and one contact socket assembly) to the size 23 wire provided (4.1.2), using the procedure of 4.4.1. The unstripped end of the wire is Inserted into the contact until it bottoms fully.

(b) Strip the insulatlon from the wire ends, leaving a minimun of 3/4 inch of insulation on each, as measureed from. the contact end.

(c) Mate 15 contact pairs.

(d) Connect the current supply and voltmeter across the stripped wire ends and increase the current until it stabilized at 1 ampere, then take the voltage drop reading immediately. The voltage drop shall conform to the requirements of 3.5.3.

(e) The 15 unmated contact pairs shall be stored unmated for 30 days on open shelf storage at room conditions. The contacts shall **then be** mated and tested as above.

4.5.4 Crimp tensile strength.

(a) The Crimp 15 contact pairs to each of the wire sizes provided (4.1.2), using the procedure of 4.4.1. Do not mate the connectors.

(b) Each contact shall remounted in the jaws of the fixed member of the tensile tester. Each crimped wire of the contact shall be connected to the wire holding cam of the moving member of the tester.

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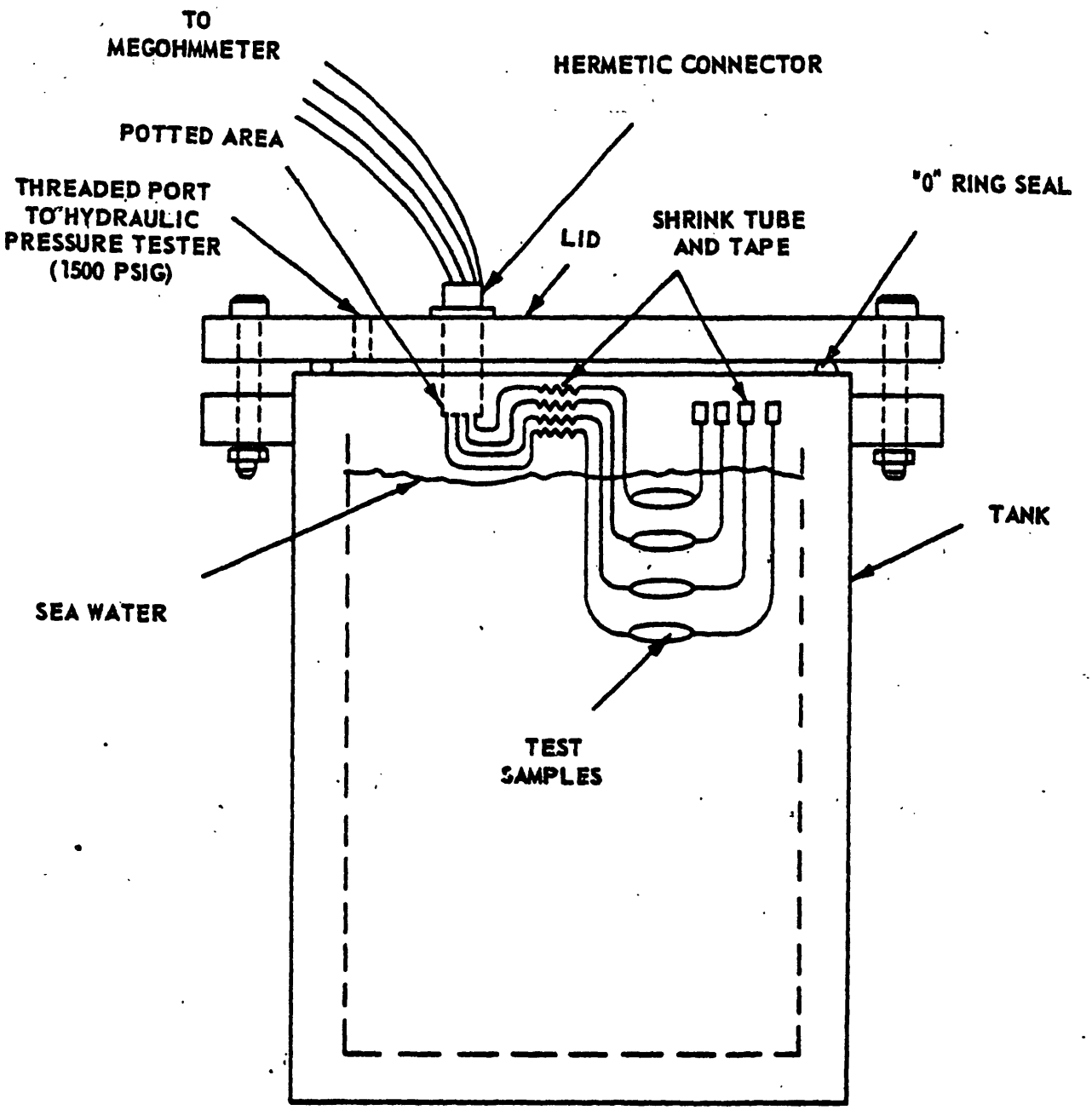


Figure 1. High Pressure Test Fixture.

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(c) A force of 26.4 ±0.2 pounds for #23 AWG or 20.8 ±0. 2 pounds for #24 AWG shall be applied for a peroid *of* not less than 30 seconds,

(d) This force shall be applied with 100 milliamps flowing through the test sample and a discontinuity monitor connected to check for any circuit opens which might occur. The connector shall meet the requirements of 3.5.4.

(e) The tensile force shall then be increased to destruction and the force and type of destruction noted.

4.5.5 Rejection criteria. Failure of a connector or part in the first article or quality conformance sample to successfully pass the required inspections and tests may result in rejection of the unit or the lot represented, as determined by the procuring activity.

4.6 Packaging inspection. The inspection of the preservation-packaging and interior package marking shall be in accordance with the group A and B quality conformance inspection requirements of MIL-P-116. The inspection of the packing and marking for shipment and storage shall be in accordance with the quality assurance provistions of the applicable container specification and the marking requirements of MIL-STD-129.

5. PACKAGING

5.1 Preservation-packaging Preservation-packaging shall be Level A.

5.1.1 Level A. The wire splice kits shall be packaged as follows:

(a) be each wire splice (dispenser) kit shall be packaged in accordance with Method 1A-8 of ML-P-116 and in accordance with Dwg 5545977-1.

(b) One each wire splice (torpedo) kit shall be ackaged in accordance with Method 1A-8 of MIL-P-116 and in accordance with Dwg 555977-2.

5.2 Packing. Packing shall be Level A.

5.2.1 Level A. The wire splice kits shall be packed as follows:

(a) Twenty-five. wire splice (dispenser) kits packaged in accordance with 5.1.1. (a) shall be packed into a W5C fiberboard box accordance with PPP-B-636. and sealed with tape conforming to PPP-T-76.

(b) Twenty-five wire splice (torpedo) kits packaged in accordance with 5.1.1 (b) shall be packed into" a W5C fiberboard box in accordance with PPP-B-636 and sealed with tape conforming to PPP-T-76.

5.2.1.1 Cushioning The unit pack, if required, shall incorporate sufficient Cushioning material to insure adequate protection from damage during shipment. At the time of packing, the moisure content of the cushioning material shall be no more than 5 percent.

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5.3 Marking

5.3.1 Special marking. The wire splice kits, packed in accordance with 5.2.1 (a) and (b), shall indicate the color of the housing on the box.

5.3.2 Normal marking. In addition to the markings required by the contract or order, unit packages and shipping containers shall be marked in accordance with the requirements of MIL-STD-129.

6. NOTES AND CONCLUDING MATERIAL

6.1 Intended use. Connectors covered by this specification are Intended for use with wire-guided torpedoes.

6.2 Ordering data

6.2.1 Procurement requirements. Procurement documents should specify the following:

(a) Title, number, and date of this specification.

(b) Assigned activity for first article and quality conformance Inspection testing.

6.2.2 Contract data requirements. When this specification is used for procure-merit which incorporates a DD Form 1423 and invokes the provisions of 7-104.9(n) of the Defense Acquisition Regulation (DAR), the data requirements identified below will be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved Contract Data Requirements List (DD Form 1423) incorporated into the contract. When the provision of DAR 7-104.9(n) are not invoked, the data specified below will be delivered by the contractor in accordance with the contract requirements. Deliverable data required by this specification is cited in the following paragraphs:

Paragraph	Data Requirement	Applicable DID
4.2.7.2 and 4.2.2.3	Reports, Test	DI-T-2072

(Copies of data item descriptions required by the contractor in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

6.3 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

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5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Code)* – Optional
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